Q24: Evaluating Agents

Outcomes, trajectories, costs, and constraints

DLS 2026

October 2026

The question

Q24
How do we judge whether an agent is good: benchmarks and metrics?

Agent evaluation is trajectory evaluation

- Observe $\rightarrow$ act/tool-call $\rightarrow$ observe $\rightarrow$ outcome
- Final answer alone misses permissions, retries, and recovery.
- Match the unit of evaluation to the deployment responsibility.

Start with a task contract

- Success condition and end-state checker
- Allowed tools and forbidden actions
- Budget: time, cost, attempts
- Human escalation and safety constraints

A scorecard, not one score

- Completion / conditional success
- Cost and wall-clock time
- Human intervention rate
- Constraint-violation rate
- Failure severity and task slices

Why success rate can mislead

- Two agents can both solve 70%.
- One may be 10x costlier or violate permissions.
- Report distributions and inspect trajectories.

Validity threats

- Benchmark contamination or leaked answers
- Brittle checks and proxy gaming
- Environment/tool version drift
- Repeated tuning on the test set

Human and automatic judges

- Human rubrics need instructions and adjudication.
- Automatic judges need validation and injection defenses.
- Preference, correctness, and safety are separate dimensions.

Takeaway

Evaluation principle
Declare the task contract; measure end state, trajectory, cost, reliability, and constraints; preserve a holdout.